

Lecture #99: Valvular Heart Disease - Clinical Findings & Diagnosis

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

Noah Chernik
Academic Medicine Scholar
nchernik@nyit.edu

Dr. Patricia Happel, DO, FACOFP
Associate Professor, Department of Family Medicine
phappel@nyit.edu



Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Learning Objectives

1. Understand the cardiac cycle and what causes the normal heart sounds.
2. Identify normal heart sounds and understand where to auscultate for them.
3. Distinguish between regurgitant and stenotic valvular lesions during systole and diastole.
4. Understand the clinical findings and patient presentations by integrating the pathophysiology of underlying diseases.
5. Recognize the appropriate steps of management for patients with valvular heart disease.

Lecture Layout

- Normal Heart Sounds
 - Understanding normal is essential to realize what is a pathology
- Abnormal Heart Sounds
 - S3 & S4 sounds
- Systolic Murmurs
 - Aortic/Pulmonic Stenosis
 - Mitral Regurgitation /Prolapse
 - Tricuspid Regurgitation
 - Ventricular Septal Defect
- Diastolic Murmurs
 - Aortic/Pulmonic Regurgitation
 - Mitral/Tricuspid Stenosis
- Physical Exam Maneuvers

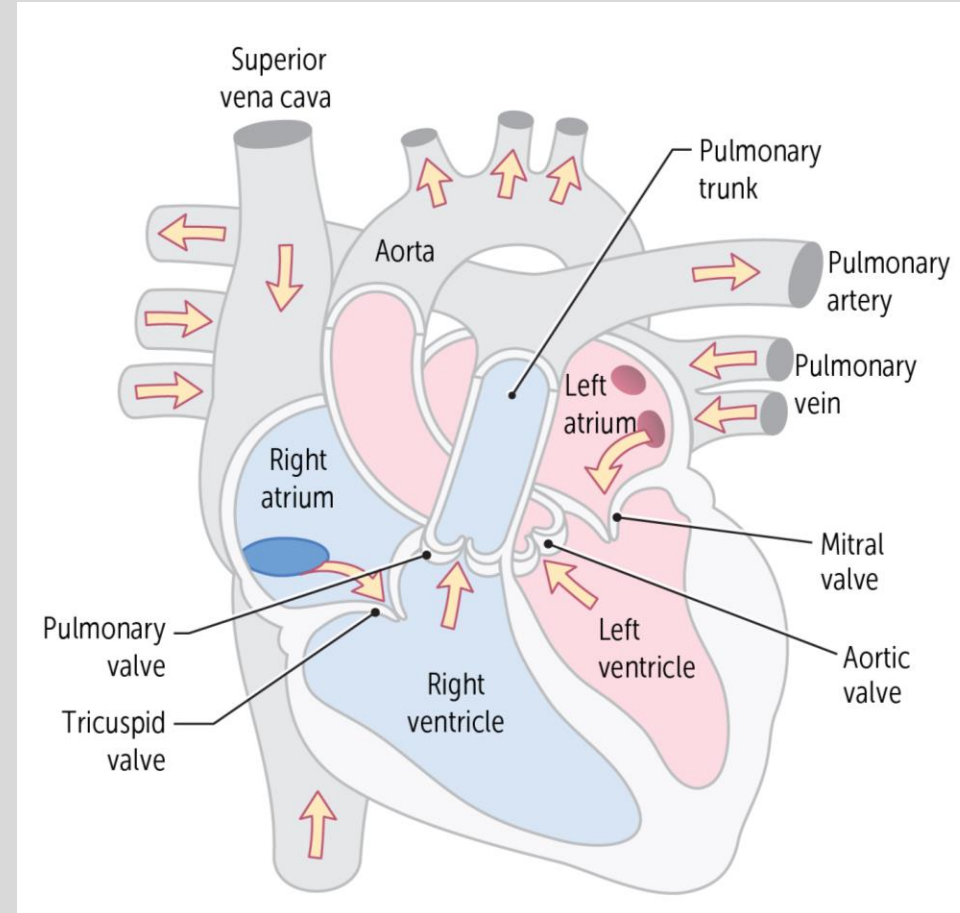
Cardiac Cycle

Systole

- Ventricular contraction
- Aortic and pulmonic valves OPEN
- Tricuspid and mitral valves CLOSE (**S₁** Heart Sound)

Diastole

- Atrial contraction
- Aortic and pulmonic valves CLOSE (**S₂** Heart Sound)
- Tricuspid and mitral valves OPEN



Physiologic Heart Sounds

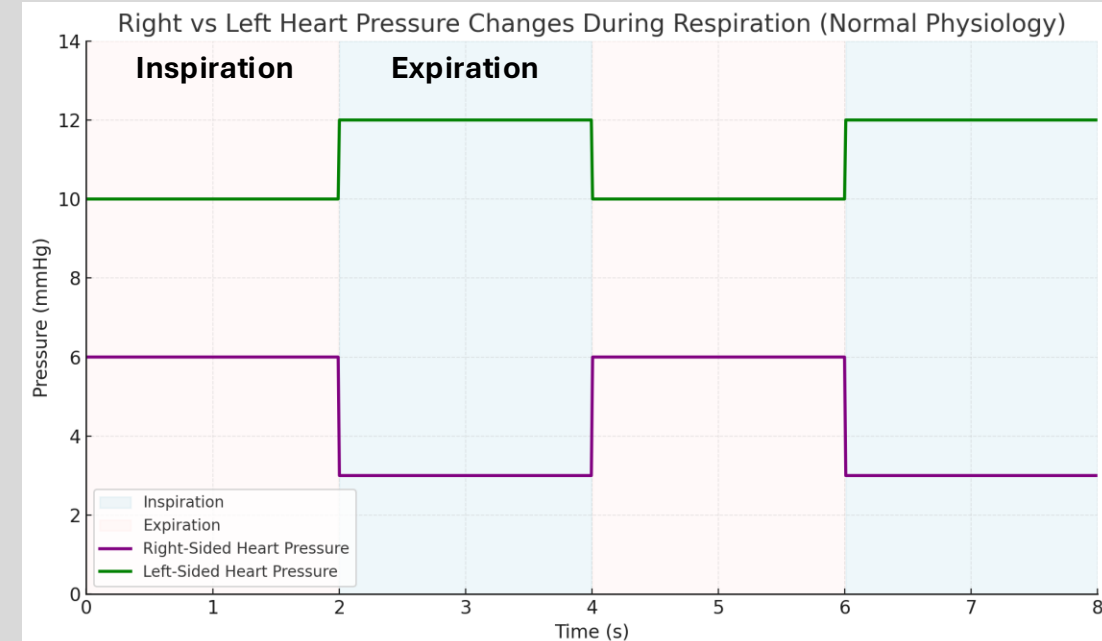
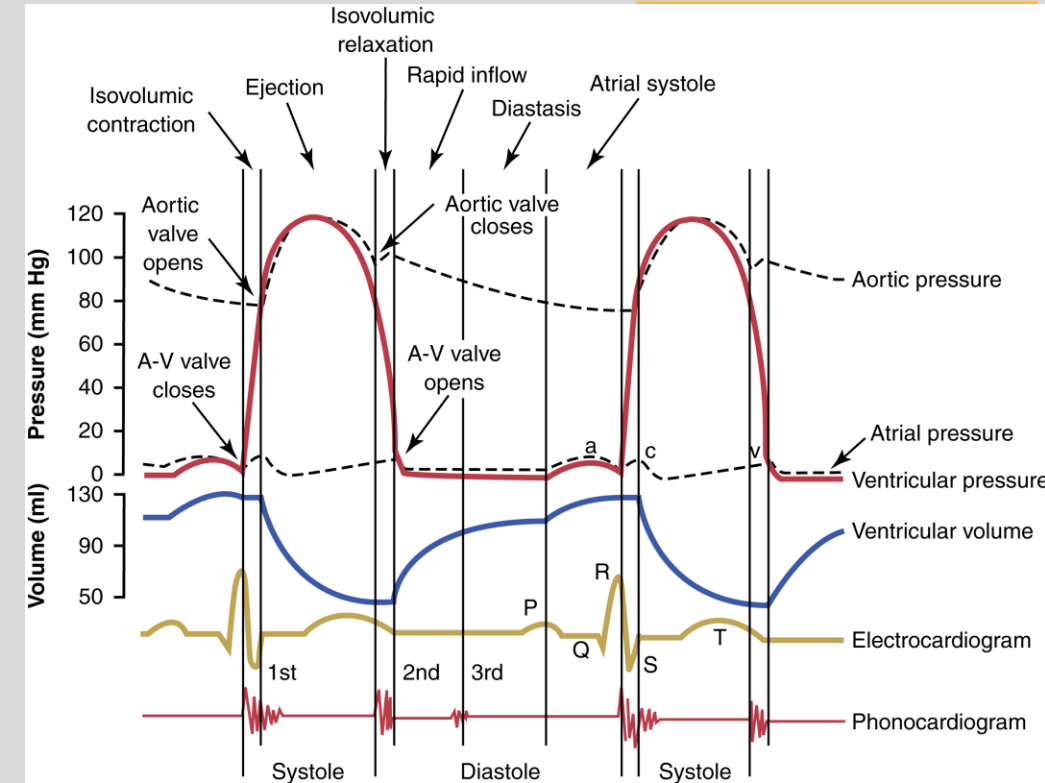


S1 (*Lub*):

- Due to the closure of the mitral and tricuspid valves
- Heard at the beginning of systole
- Louder at apex
- Mechanism: increased ventricular pressure closes the valves

S2 (*Dub*):

- Due to the closure of the aortic and pulmonic valves
- Splitting of S2
 - Inspiration: increased splitting between the closure of the aortic and pulmonic valves
 - Inspiration will **INCREASE** the amount of preload in the right ventricle, which will take a longer time to equilibrate
 - **P2** will occur **AFTER** A2, as there is a larger pressure differential between the RV and the pulmonic arteries
 - **** Reminder:** valves close when pressures are equal
- Louder at base
- Mechanism: decreased ventricular pressure closes the valves



Areas of Auscultation

Five Main Areas for Auscultation of the Heart:

- **Aortic:** right 2nd intercostal space
- **Pulmonic:** left 2nd intercostal space
- **Tricuspid:** left lower sternal border at 4th intercostal space
- **Mitral:** left 5th intercostal space
- **Erb's Point:** left 3rd intercostal space

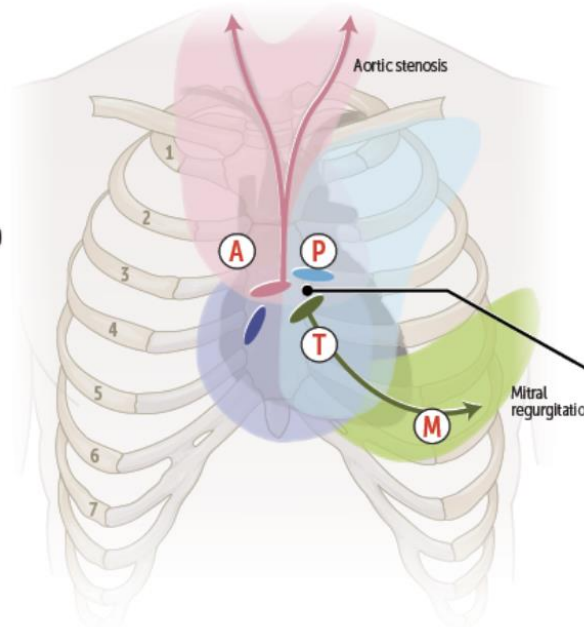
Auscultation of the heart

Where to listen: **APT M**

A Aortic area:
Systolic murmur
Aortic stenosis
Flow murmur
(eg. physiologic murmur)
Aortic valve sclerosis

T Tricuspid area:
Holosystolic murmur
Tricuspid regurgitation
Ventricular septal defect
Diastolic murmur
Tricuspid stenosis

● Aortic
● Pulmonic
● Tricuspid
● Mitral



P Pulmonic area:
Systolic ejection murmur
Pulmonic stenosis
Atrial septal defect
Flow murmur

M Mitral area (apex):
Systolic murmur
Mitral regurgitation
Mitral valve prolapse
Diastolic murmur
Mitral stenosis

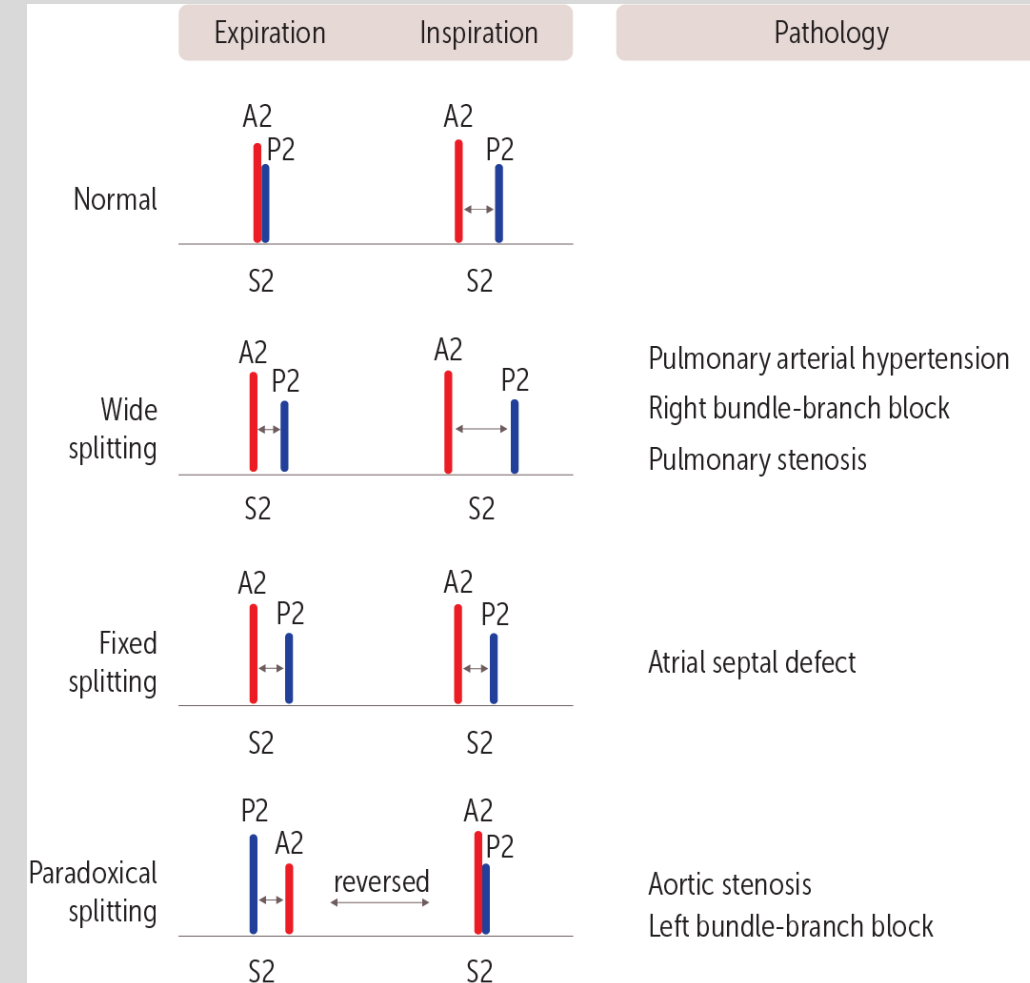
Left sternal border:
Systolic murmur
Hypertrophic
cardiomyopathy
Diastolic murmur
Aortic regurgitation
Pulmonic regurgitation

Splitting of S2



Splitting of S2: A2 heard before P2

- Occurs during inspiration
 - The aortic valve will close **BEFORE** the pulmonic valve during inspiration
 - Best heard in the pulmonic area
- Mechanism:
 - Inspiration will **DECREASE** intrathoracic pressure, allowing for increased volumes of blood to fill the right ventricle (RV)
 - Increase of preload in the RV creates a larger pressure difference between the RV and the pulmonic artery
 - During systole, the pulmonic valve will stay open longer since the pressure gradient is wider
 - ** Reminder:** valves close only when the pressure on either side of the valve is equal
- Can be pathologic:
 - Atrial Septal Defect: Fixed splitting (does not disappear during expiration)
 - Aortic Stenosis/Left Bundle Branch Block: Paradoxical splitting of S2 occurs during exhalation
 - Pulmonary Arterial Hypertension/Stenosis: wide splitting



Splitting of S2

Can be pathologic:

Fixed Splitting: does not disappear during expiration

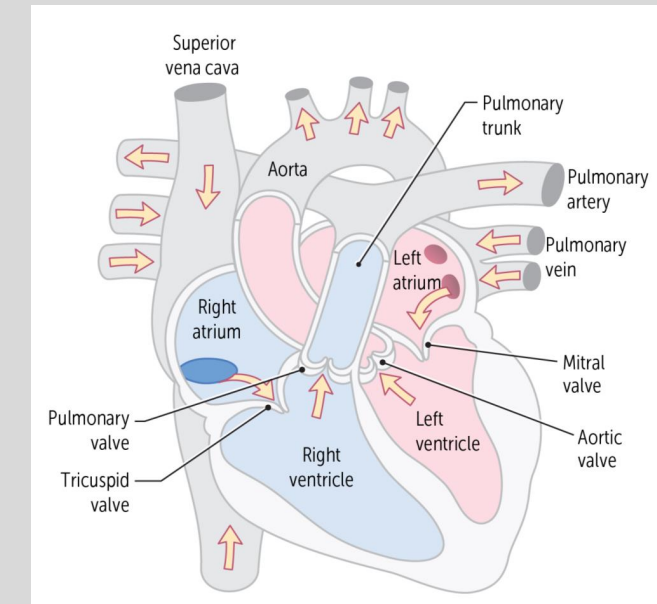
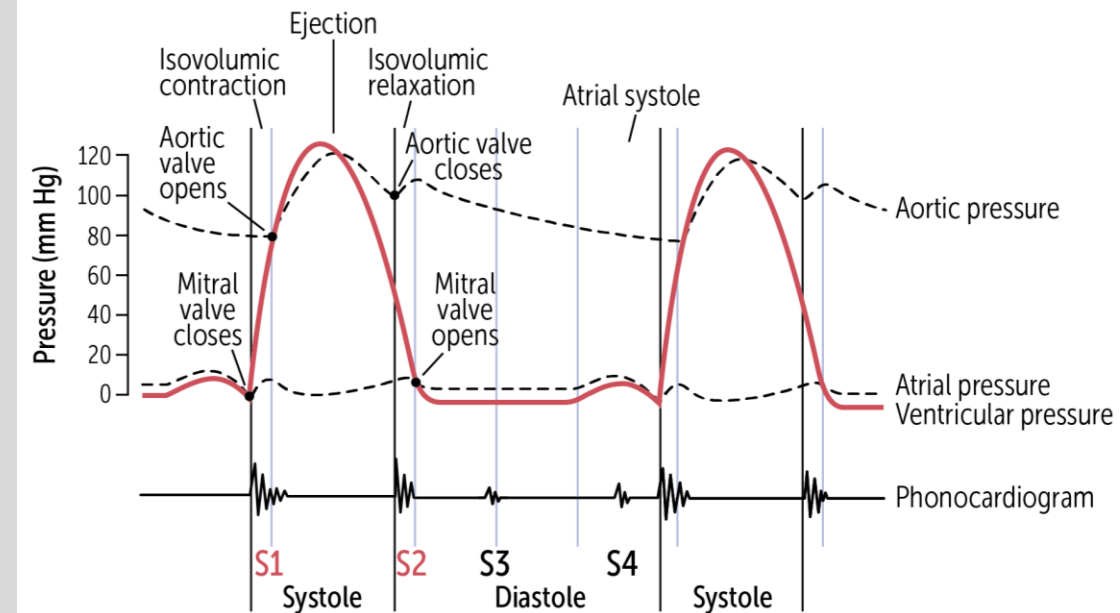
- **Atrial Septal Defect**
 - Right atrium has lower pressure than the left, which allows for blood from the LA to rush into the RA.
 - Excess blood is then pumped into the RV, increasing the preload regardless of inspiration

Paradoxical splitting of S2 occurs during exhalation

- Aortic Stenosis: valve is stiff, takes longer to close

Wide splitting: normal S2 split is exaggerated

- **Pulmonary Arterial Hypertension/Stenosis:**
 - PAH: Increased pulmonary artery pressure increases the duration of systole of the RV, since the RV must overcome increase pressure gradients to open the pulmonic valve
 - Stenosis: valve is stiff, and takes longer to close



S3 & S4



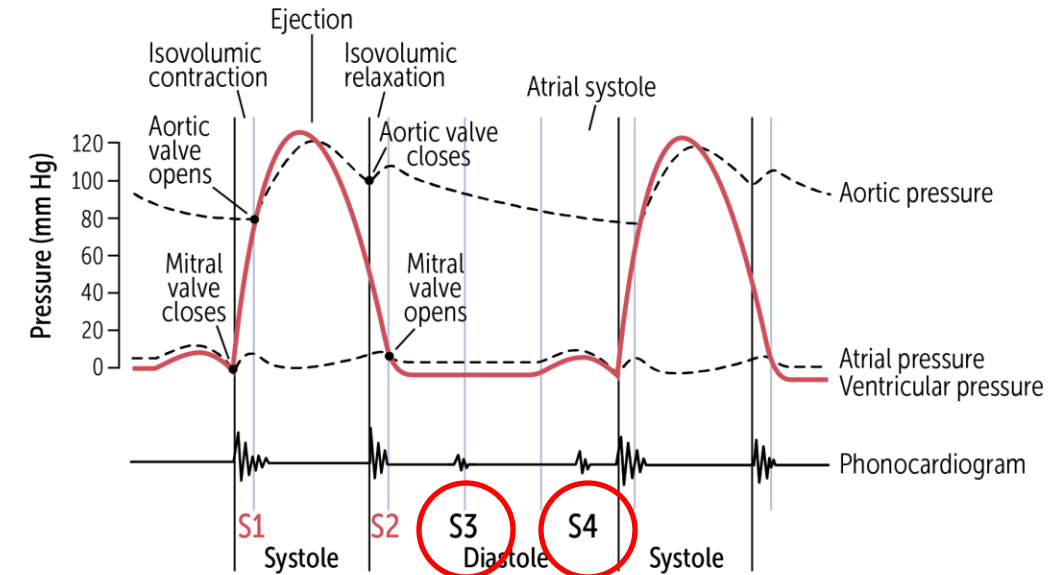
S3: volume overload

- Occurs immediately after S2
- Sound is caused by blood rushing into an already filled (volume overloaded chamber)
- **Normal** in well conditioned athletes, children, and pregnant women
- **Abnormal:** think about aortic valve regurgitation, dilated cardiomyopathy



S4: compliance issue

- Occurs right before S1
- Sound is caused by blood hitting stiff ventricular walls



Murmur

Typically, a murmur can be broken down into 1 of 2 causes:

- Stenosis: problem with Opening the valve
 - Also known as a forward flowing murmur
 - Pathophysiology Connection: when a valve does not open properly/fully, the heart chamber “upstream” of the stenotic valve must **increase** force product
 - This causes hypertrophy and thickening of the chamber
- Regurgitation: problem with closing the valve
 - Also known as a backwards flowing murmur
 - Pathophysiology Connection: when a valve cannot close properly, the heart chamber “upstream” of the regurgitation will experience increase blood and volumes
 - This causes dilation of the chamber

Describing Murmurs

Sound:

Crescendo Decrescendo: rises, then falls in intensity

- Examples: aortic stenosis

Plateau: same intensity throughout

- Examples: mitral regurgitation

Crescendo: grows louder than disappears

- Example: mitral stenosis

Decrescendo: grows softer

- Example: aortic regurgitation

Location:

- Louder at the base: aortic
- Louder at the apex: mitral

Radiation:

- The sound can radiate into the carotids (aortic stenosis) or axilla (mitral regurgitation)

Quality: Blowing, harsh, rumbling

Pitch: low, medium, high

Grading Murmurs

Grade	Description
Grade 1	Very faint, heard only after listener has “tuned in”; may not be heard in all positions
Grade 2	Quiet, but heard immediately after placing the stethoscope on the chest
Grade 3	Moderately loud
Grade 4	Loud, with palpable thrill
Grade 5	Very loud, with thrill. May be heard when the stethoscope is partly off the chest
Grade 6	Very loud, with thrill. May be heard with stethoscope entirely off the chest

Dynamic Auscultation

Respiration:

- **Inhalation:** increases the intensity of any right sided murmurs ($\uparrow$ RV preload & $\downarrow$ LV preload)
- **Expiration:** increases the intensity of any left sided murmurs ($\uparrow$ LV preload)

Valsalva Maneuver: patient will “bear down” by attempting to exhale with a closed mouth

- RV Preload, LV Preload, LV Afterload will **DECREASE**
- Key use: distinguish aortic stenosis vs hypertrophic obstructive cardiomyopathy (worsens HOCM)

Squatting from Standing & Leg Raising: blood stored in the legs will return to the heart

- RV and LV Preload will **INCREASE**, no change in LV afterload
- Keys: soften the murmur of HOCM and move the click in mitral valve prolapse **later** in systole

Standing from Seated/Squatting: blood will pool in the legs

- Results in **DECREASED PRELOAD**
- Similar to Valsalva maneuver
- Keys: move the click in mitral valve prolapse **earlier** in systole

Handgrip:

- Results in **INCREASED AFTERLOAD**
- Keys: increase the intensity of left-sided regurgitant (backward flowing) murmurs

Murmur Categories

Systolic	Diastolic
Aortic Stenosis	Aortic Regurgitation
Pulmonic Stenosis	Pulmonic Regurgitation
Mitral Regurgitation	Mitral Stenosis
Tricuspid Regurgitation	Tricuspid Stenosis
Mitral Valve Prolapse	

Systolic Murmurs

Aortic Stenosis



Causes

- Post-inflammatory scarring from rheumatic heart disease
- **Senile calcified aortic stenosis** (occurs with aging)
- **Calcification of congenitally deformed valve** (bicuspid valve, **occurs at a younger age**)

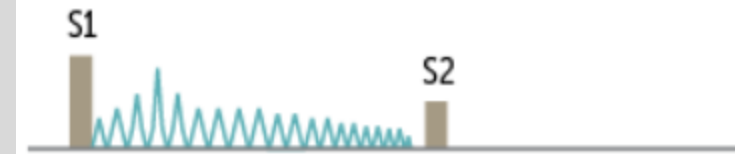
Murmur Description:

- Crescendo-decrescendo systolic murmur
- Heard at the aortic area of auscultation (right 2nd intercostal space)
- Radiates to the carotids

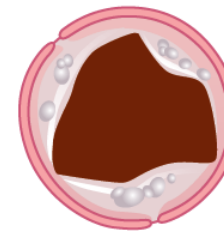
Dynamic Auscultation (forward flow murmur)

- Standing/Valsalva: decreased intensity (↓ preload)
- Handgrip: decreased intensity (↑ afterload)
- Squatting: increased intensity (↑ preload)
- Leg Raise: increased intensity (↑ preload)

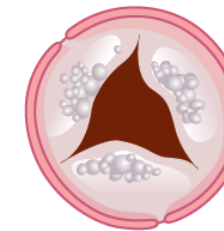
Aortic stenosis



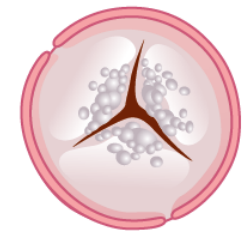
Normal



Aortic sclerosis



Mild to moderate
aortic stenosis



Severe aortic stenosis

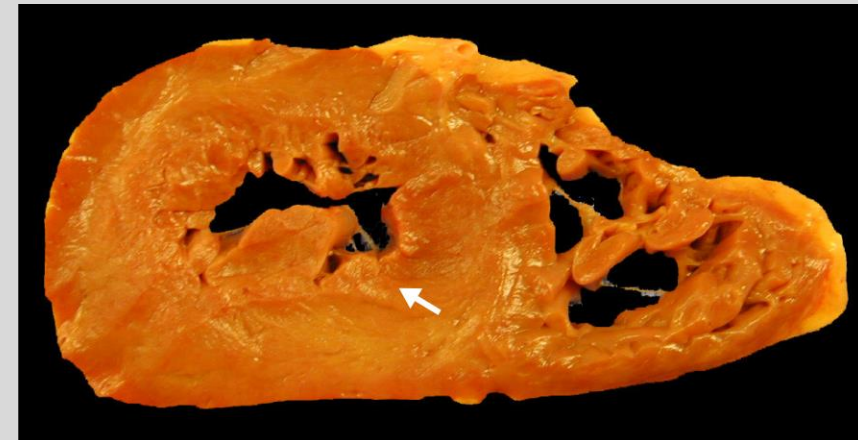
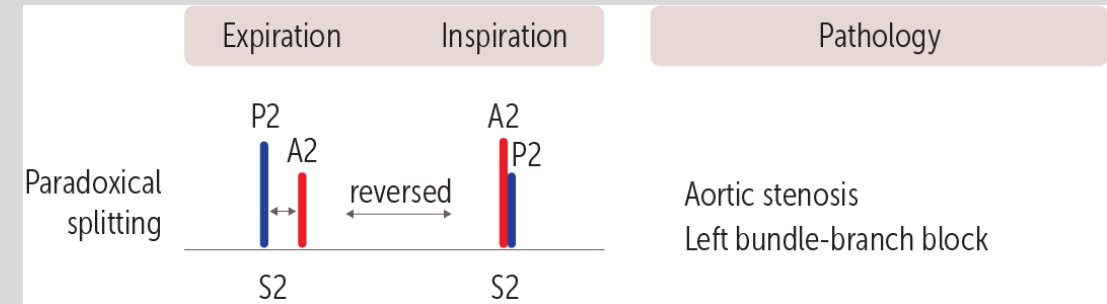
Aortic Stenosis

Clinical Presentation

- Can cause **paradoxical splitting of S2**, as the aortic valve takes longer to open as the disease progresses
- **Syncope, angina, dyspnea** on exertion
 - Respiratory crackles
- “Pulsus parvus et tardus”: weak and delayed pulse in the carotids
- S4 heart sound may become present, as blood rushes in against stiff, hypertrophied ventricular walls
- Larger and more lateral point of maximal impulse

Pathologic Changes to the Heart

- Left ventricular hypertrophy as an adaptive measure to counteract the increased force required to eject blood from the LV



LVH

Pulmonic Stenosis



Causes:

- Congenital (Teratology of Fallot)
- Carcinoid syndrome (affects both tricuspid and pulmonary valves)
- Endocarditis

Murmur Description:

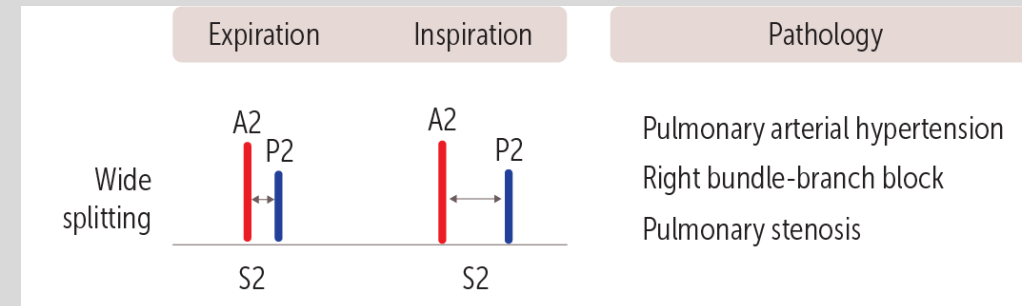
- **Crescendo-decrescendo mid-systolic high-pitched murmur**
- Heart at the pulmonic area of auscultation (left 2nd intercostal space near the sternal border)
- Radiates to the left shoulder and neck

Dynamic Auscultation (forward flow murmur)

- Increased intensity with **INCREASED preload**
- Increased intensity with **inspiration**
- Decreased intensity with **DECREASED preload**

Clinical Presentation:

- Can cause wide splitting of S2, as the RV takes longer to eject blood through the calcified pulmonic valve
- Anginal chest pain



Mitral Regurgitation

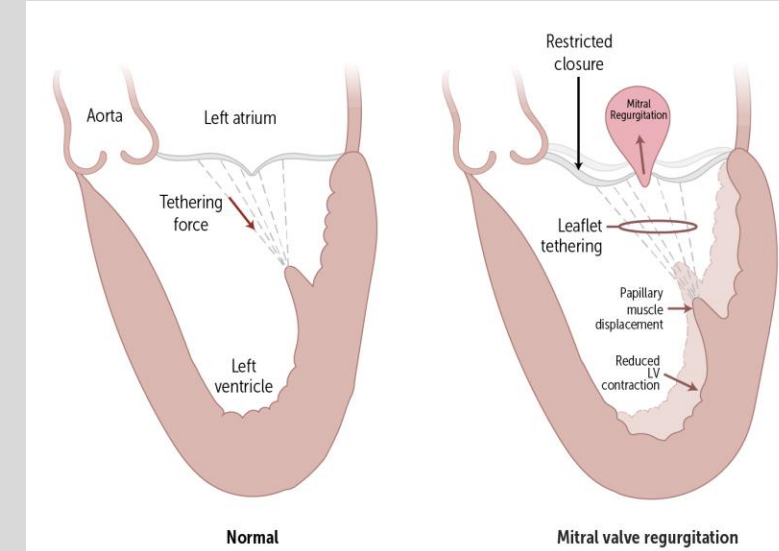


Causes: **mitral valve prolapse** can precede mitral regurgitation

- Post myocardial infarction: acute papillary muscle rupture
- Chronic Infective Endocarditis: vegetation and valve deformities on the leaflets of the valves
- Rheumatic heart disease: inflammation and damage to mitral valve
- Congenital Deformities:
 - Diseases that weaken the collagen of the valve (Marfan syndrome, Ehlers-Danlos syndrome, osteogenesis imperfecta)
- Ventricular Disease:
 - Dilated Cardiomyopathy: the opening of the mitral valve is too wide for the leaflets, allowing blood to pass through
 - Hypertrophic Cardiomyopathy: anterior leaflet is pulled towards the thickened wall, allowing blood to pass through

Murmur Description:

- Blowing holosystolic murmur
- Best heard at apex of the heart (left 5th intercostal space at midclavicular line)
- Usually obscures S1 (the valve does not close properly!)



Mitral Regurgitation

Dynamic Auscultation:

- Will **decrease** in intensity during **inspiration** (left sided valvular pathology)
- **INCREASE** in intensity with **INCREASED** preload and afterload
 - Preload: more blood in the LV, more has the opportunity to escape back through the mitral valve
 - Afterload: increased resistance for blood leaving the aortic valve will increase the pressure of the LV, causing more blood to pass back through the mitral valve
- **DECREASE** in intensity with **DECREASED** preload

Clinical Presentation:

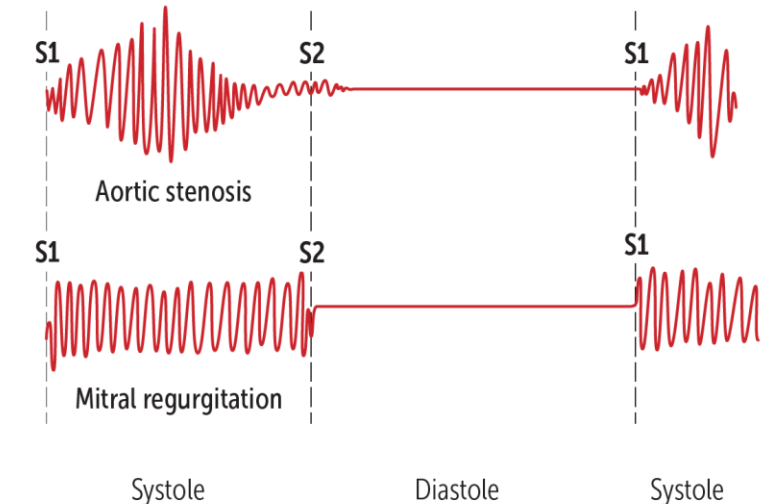
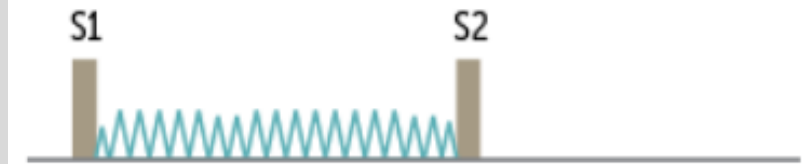
- Pulmonary crackles
- Elevated jugular venous distension
- Lower extremity edema

Pathologic Changes to the Heart:

- Preload in the LA will increase, as blood moves backwards from the LV
- During diastole (atria contract), this extra blood will move back into the LV, **increasing** its end diastolic pressure
- This backup continues from the LA, as the pulmonary veins become congested
 - Increased pulmonary capillary pressure and pulmonary edema (pillow orthopnea)

- "Mitral Regurgitation." Harrison's Principles of Internal Medicine, 22nd Edition
- "Approach to the Patient with a Heart Murmur." Harrison's Principles of Internal Medicine, 22nd Edition
- "Mitral Regurgitation." ScholarRX Brick
- "Heart Murmurs." FirstAid 2025

Mitral/tricuspid regurgitation



Mitral Valve Prolapse

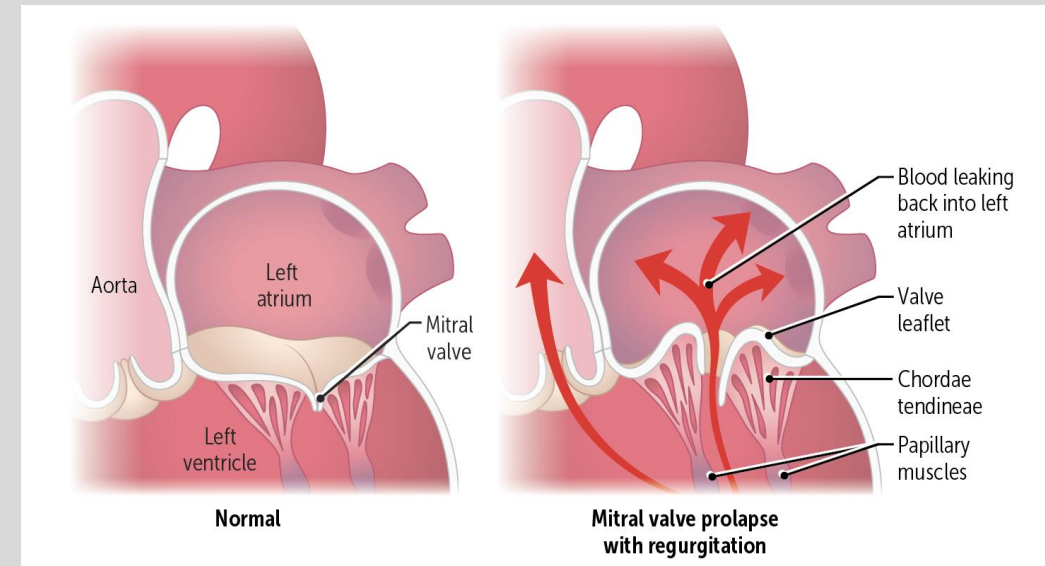


Pathophysiology:

- The leaflets are attached to the ventricle via the chordae tendineae with the help of papillary muscles
- In MVP, the leaflets and chordae tendineae weaken, allowing the leaflets to balloon out into the LA during systole

Causes:

- Myxomatous degeneration: irregular middle layer (spongiosa) of the valve leaflet
 - Leaflet thickens and chordae tendineae weaken
 - Increased deposition of mucopolysaccharides in the spongiosa layer (middle layer) of the valve
- Connective tissue disease: Marfan syndrome, EDS, osteogenesis imperfecta
 - **Marfan**: AD defective fibrillin production, which crosslinks elastin
- Mitral valve disease
 - Infective Endocarditis: scar leaflets and weaken chordae tendineae
 - Post MI: tearing of the papillary muscles
 - Rheumatic Fever: inflammation allowing the leaflets to stretch



Mitral Valve Prolapse

Murmur Description:

- Mid-to-late systolic click
 - The click is the mitral chordae tendineae snapping as the leaflet prolapses into the LA

Dynamic Auscultation: click will change with LV preload

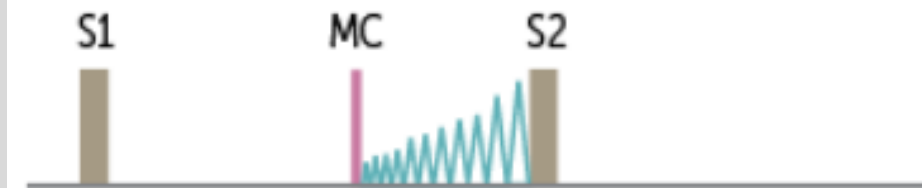
- Decreased preload: earlier click
- Increased preload: later click
- Think about the “click” occurring at some predetermined volume that is less than LVEDP
 - More preload or higher afterload, click will occur later in systole as more fluid must leave the heart to achieve the “click” volume

Clinical Presentation

Pathologic Changes to the Heart

- Can eventually lead to mitral regurgitation

Mitral valve prolapse



Tricuspid Regurgitation



Pathophysiology

- Due to RV dilation not allowing the tricuspid valve to fully close during systole

Causes

- Due to RV dilation (pulmonary hypertension)
- **Ebstein's anomaly:**
 - Downward displacement of the tricuspid valve into the RV
 - Due to Lithium in utero
- **Infective endocarditis:** IV drug user (first valve *Staph aureus* encounters in the heart)

Murmur Description:

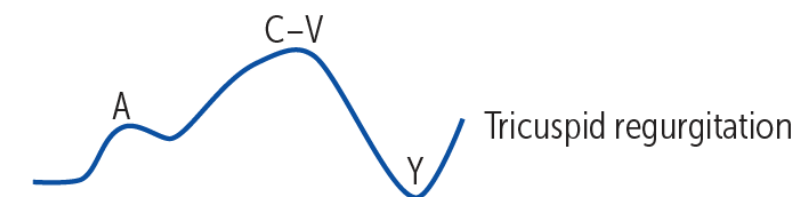
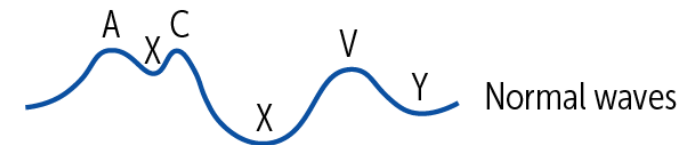
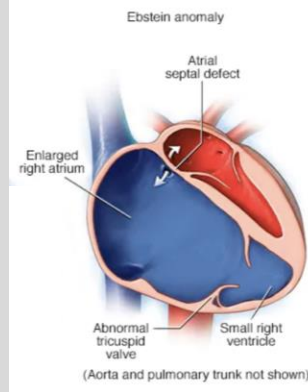
- Holosystolic high-pitched “blowing murmur”
- Loudest in tricuspid area
- Barely audible S1 with split S2

Dynamic Auscultation

- Increases with inspiration

Clinical Presentation

- Typically asymptomatic, severe cases may present with symptoms of right heart failure
- Creates changes in jugular venous waveform (giant c-v waves during systole)



Diastolic Murmurs

Aortic Regurgitation

Pathophysiology

- Blood rushes back through the aortic valve into the LV, known as the regurgitant fraction. This causes a drop in diastolic pressure
- Regurgitant fraction also must be ejected during systole, increasing systolic pressure
- Will see an elevated LV EDP

Causes

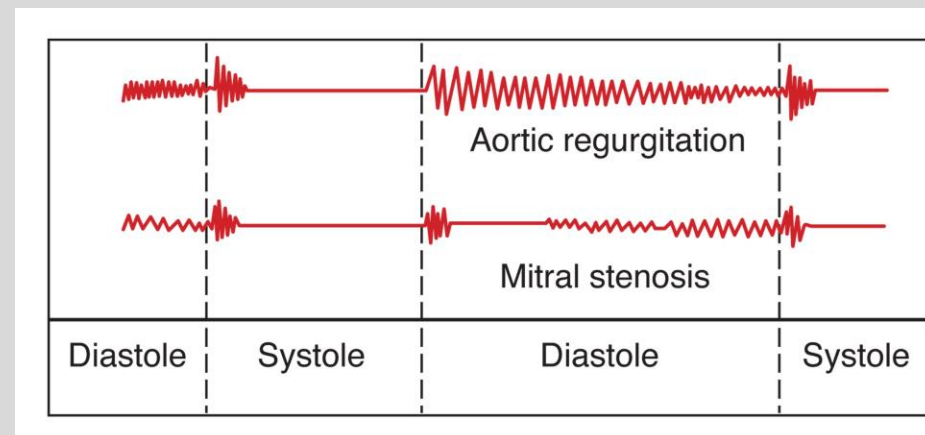
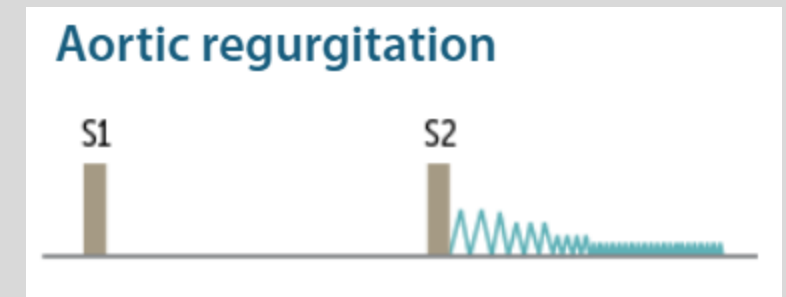
- Bicuspid aortic valve
- Endocarditis
- Aortic root dilation
- Rheumatic fever
- Hypertension

Murmur Description:

- Early diastolic, decrescendo high-pitched “blowing” murmur
- Aortic Root Dilation: base (right upper sternal border)
- Valvular Pathology: left sternal border (left 3rd intercostal space)

Dynamic Auscultation

- Louder during expiration
- Patient sitting up and leaning forward



- “Aortic Regurgitation.” Harrison’s Principles of Internal Medicine, 22nd Edition
- “Approach to the Patient with a Heart Murmur.” Harrison’s Principles of Internal Medicine, 22nd Edition
- “Aortic Regurgitation.” ScholarRX Brick
- “Heart Murmurs.” FirstAid 2025

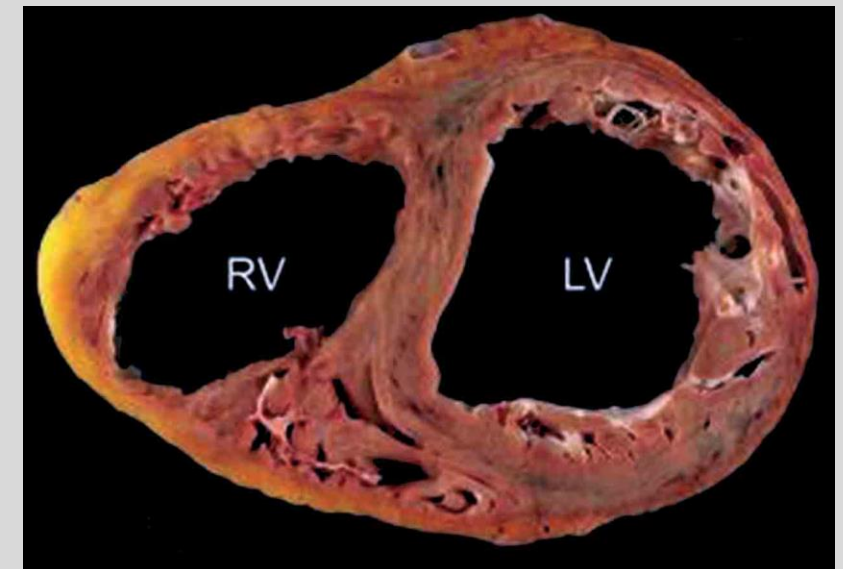
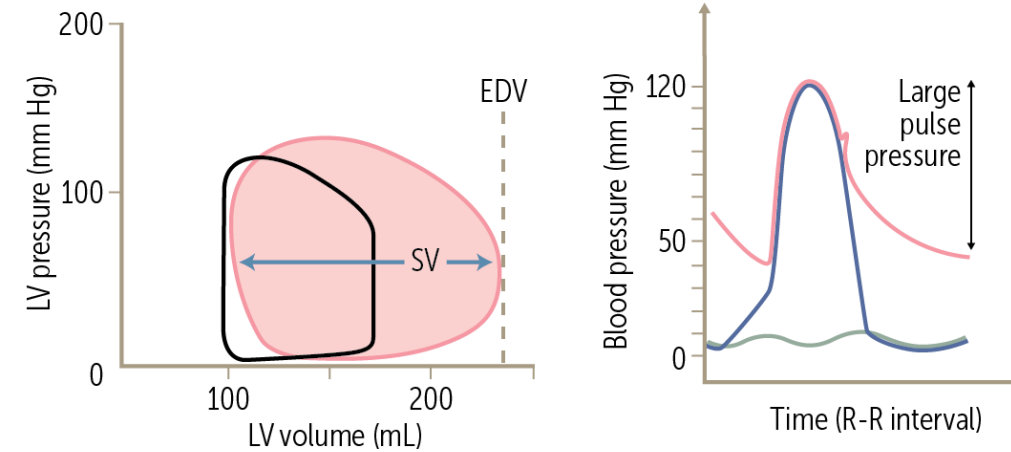
Aortic Regurgitation

Clinical Presentation

- Hyperdynamic pulse (large difference between systolic and diastolic BPs)
- Head bobbing
- Pillow orthopnea/paroxysmal nocturnal dyspnea
- S3 due to dilated cardiomyopathy

Pathologic Changes to the Heart

- Can progress to left sided HF
- To manage the increased end diastolic pressures and force required to achieve systole, the LV will hypertrophy, causing eccentric hypertrophy
 - As the LV EDP continues to increase, the LV will dilate, causing **dilated cardiomyopathy**
- Causes heart failure: congestion will move “backwards” throughout the system, leading to pulmonary complications (edema, dyspnea, etc)



Pulmonary Regurgitation

Causes: uncommon

- Can be seen post-op for pulmonary stenosis
- Pulmonary hypertension
- Infective endocarditis

Murmur Description:

- High-pitched, early decrescendo murmur
- Heard at Erb's point (left 3rd or 4th intercostal space)
- Description: normal S1 with P2 later than A2

Dynamic Auscultation

- Increases in intensity during inspiration (helps differentiate from AR)

Clinical Presentation

- Palpable P2

Mitral Stenosis

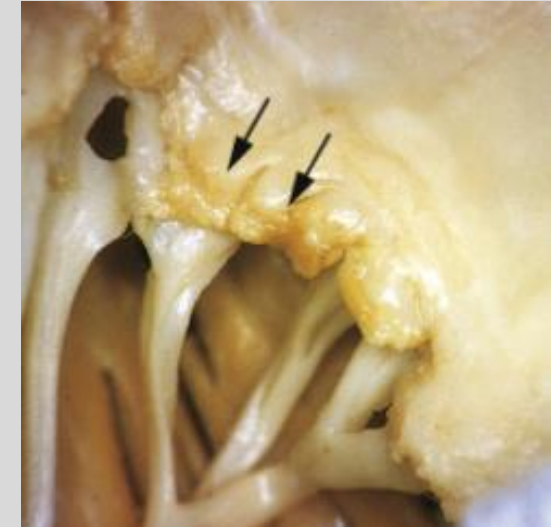


Pathophysiology

- Narrowing/hardening of the mitral valve
- Will see increase LA volume

Causes

- Most common cause is rheumatic heart disease
 - **Complication of acute rheumatic fever** due to GAS (Group A Strep/Strep Pyogenes)
 - Typically a **longer term process**, as inflammation will slowly calcify the valve (chronic presentation)
 - “Fish mouth” presentation of the mitral valve
- Calcification: age related calcium deposits
- Radiation: Hodgkin-lymphoma or chest malignancies that receive extensive radiation



Murmur Description:

- Opening snap followed by a low-pitched diastolic rumble
 - Opening snap is due to an abrupt stop of the leaflet tips due to fusion
- Best heard at apex of the heart
- As disease severity increases, the opening snap occurs earlier in diastole
 - This is due to increasing LA pressure, forcing the mitral valve to “snap” open earlier in diastole

Mitral stenosis



Mitral Stenosis

Dynamic Auscultation

- Increases in intensity with expiration & when positioned in left lateral decubitus
- Decreases intensity with decreased preload

Clinical Presentation

- Exertional dyspnea
- Pillow orthopnea/paroxysmal nocturnal dyspnea

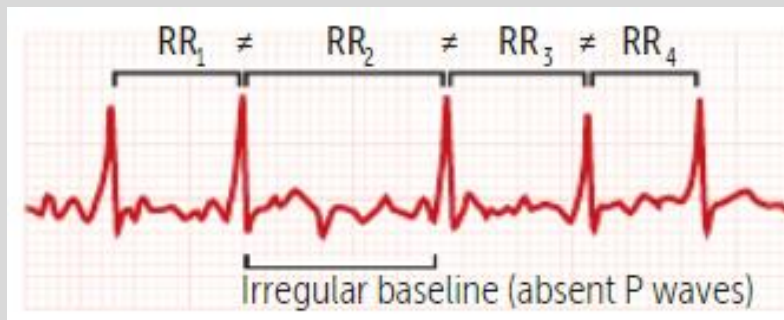
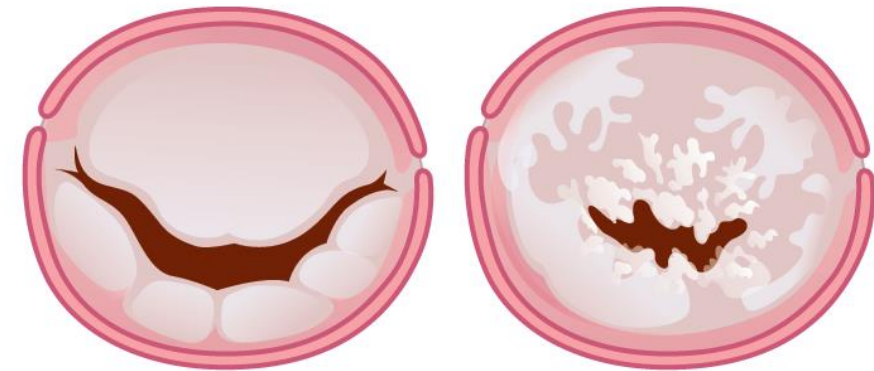
Pathologic Changes to the Heart

- Increased LA volume will cause pulmonary edema and congestion
- Can cause atrial fibrillation due to LA dilation
- Thromboembolism: LA dilation can allow for slow moving blood to pool in the LA

Normal mitral valve

Calcified mitral valve

(View from below, looking up)



Tricuspid Stenosis

Pathophysiology

- Inflammation or chronic injury causes fibrosis and calcium deposition, leading to a thickened and stiff valve
- Valve will not open fully during diastole
- Increases pressures in RA and vena cava

Causes

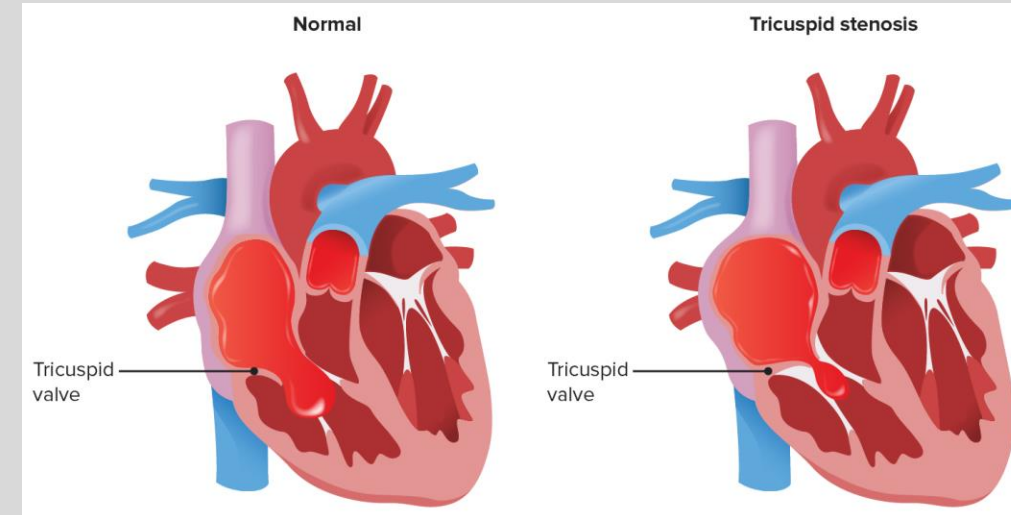
- Rheumatic heart disease
- Atrial myxoma: tumor can block the valve
- Nonbacterial thrombotic endocarditis (SLE)

Murmur Description:

- Opening snap, followed by soft mid-diastolic rumble
- Best heard at fourth intercostal space at left parasternal border

Dynamic Auscultation (forward flow murmur)

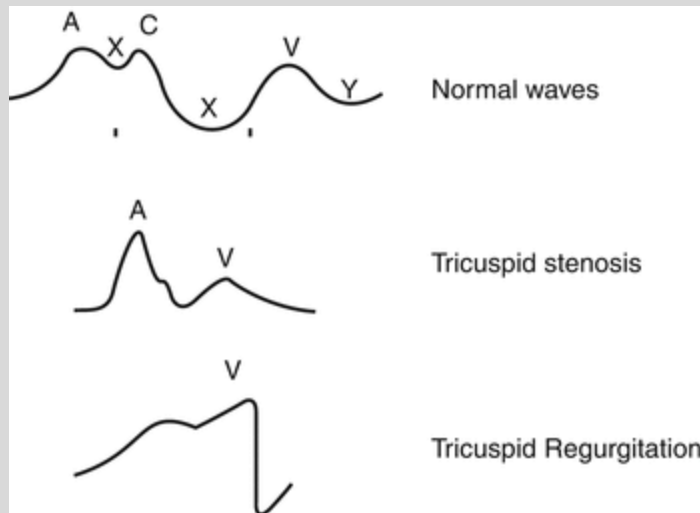
- Increased preload (inspiration, leg raising, squatting) will increase intensity
- Decreased preload (standing/Valsalva) will decrease intensity



Tricuspid Stenosis

Clinical Presentation

- Asymptomatic until disease is severe
- Palpitations, fatigue, cold skin (due to lower cardiac output)
- Left heart failure as a result of inadequate preload
- Giant *a* wave on the JV waveform, as the RA is contracting against a narrowed tricuspid valve
- Kussmaul Sign: paradoxical increase in jugular venous pressure during inspiration
 - Not specific to tricuspid stenosis, occurs in any condition where blood flow from the vena cava to the pulmonary vasculature is compromised (tricuspid regurgitation, constrictive pericarditis, restrictive cardiomyopathy)



- "Tricuspid Valve Disease." Harrison's Principles of Internal Medicine, 22nd Edition
- "Approach to the Patient with a Heart Murmur." Harrison's Principles of Internal Medicine, 22nd Edition
- "Tricuspid Regurgitation and Tricuspid Stenosis." ScholarRX Brick
- "Heart Murmurs." FirstAid 2025

Non-Valvular Murmurs

Ventricular Septal Defect

- Systolic murmur
- Holosystolic, harsh sounding murmur
- Best heard at tricuspid area



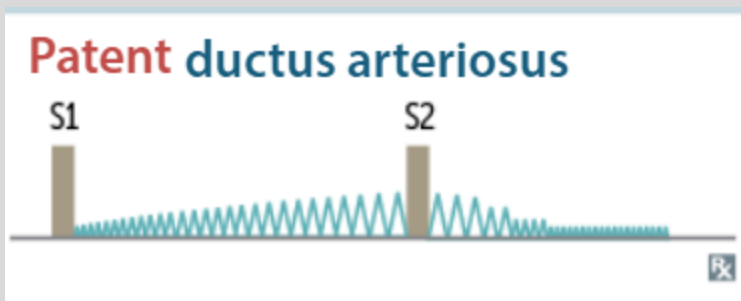
Atrial Septal Defect

- Wide-fixed splitting of S2 due to increased volume in the RA
- May see right heart dilation due to fluid overload



Patent Ductus Arteriosus

- Continuous, **machine** like murmur
- Best heard at left infraclavicular area
- Louder in S2



	Expiration	Inspiration	Pathology
Normal	A2 P2 S2	A2 P2 S2	
Wide splitting	A2 P2 S2	A2 P2 S2	Pulmonary arterial hypertension Right bundle-branch block Pulmonary stenosis
Fixed splitting	A2 P2 S2	A2 P2 S2	Atrial septal defect
Paradoxical splitting	P2 A2 S2	A2 P2 S2	Aortic stenosis Left bundle-branch block

General Rules/Considerations

Dynamic Auscultation Review:

Standing/Valsalva: decreased preload

- Decrease the fluid in the heart, should decrease most murmurs involving fluid moving across the valve (regurgitation & stenosis)
- MVP: remember the “click” occurs at a setpoint that is less than the normal end diastolic volume, so if preload is decreased this click occurs **SOONER**

Passive Leg Raise: increased preload

- Increase the fluid in the heart, should increase most murmurs involving fluid moving across the valve (regurgitation & stenosis)
- MVP: takes longer for the click to occur, as more blood must exit the LV prior to achieving the volume where the click occurs

Squatting: increased preload **AND** afterload

- Increased fluid in the heart, similar to above
- **The afterload increases, but preload is still greater**

Handgrip: increased afterload

- Usually only left sided murmurs are affected
- Afterload will **increase the effects of backward flowing murmurs** (aortic/mitral regurgitation, ventral septal defect)
- Decrease aortic stenosis

Inspiration: increase **RIGHT SIDED MURMURS**

- Look for this in question stems, helps you eliminate essentially half of the murmur options

Dynamic Auscultation: Systolic Murmurs Severity

Disease	Standing/Valsalva ↓ Preload	Leg Raise ↑ Preload	Squat ↑ Preload & Afterload	Handgrip ↑ Afterload	Inspiration ↑ Preload
Aortic Stenosis	↓	↑	↑	↓	↓
Pulmonic Stenosis	↓	↑	↑	-	↑
Mitral Regurgitation	↓	↑	↑	↑	↓
Mitral Valve Prolapse	↑	↓	↓	-	↓
Tricuspid Regurgitation	↓	↑	↑	-	↑

Dynamic Auscultation: Diastolic Murmurs Severity

Disease	Standing/Valsalva ↓ Preload	Leg Raise ↑ Preload	Squat ↑ Preload & Afterload	Handgrip ↑ Afterload	Inspiration ↑ Right Preload
Aortic Regurgitation	↓	↑	↑	↑	↓
Pulmonic Regurgitation	↓	↑	↑	-	↑
Mitral Stenosis	↓	↑	↑	↓	↓
Tricuspid Stenosis	↓	↑	↑	-	↑

Complications

Stenosis:

- Increased pressure in the chamber before the diseased valve
 - Think about hypertrophy and backflow of blood

Regurgitation:

- Increased volume in chamber before the diseased valve
 - **Think about backflow and dilation**
- **Leads to heart failure if untreated**

Left Sided Murmur/HF: will cause pulmonary congestion and edema

- **Cough and dyspnea**
- Orthopnea, paroxysmal nocturnal dyspnea (Patient must sleep in a recliner)

Right Sided Murmur/HF: engorgement of system and portal venous system

- Hepatomegaly, jugular venous distension, peripheral edema

Algorithm: Clinical

1. Is this normal?

- The first thing to identify is if the rate and rhythm you hear are within normal limits
- If not, then proceed down this checklist
- Even if you cannot identify the murmur, being able to determine what is abnormal will help your presentations during clinical rotations

2. What is the abnormal sound?

- Is it a S3 or S4 heart sound
- Is it a valvular murmur

3. When is the murmur occurring?

- Systole, diastole, continuous
- Helps narrow down your potential options

4. Where do I hear the murmur?

- Think about your areas of auscultation
- Is the murmur radiating anywhere

5. What does the murmur sound like?

- Blowing, crescendo-decrescendo, etc
- Does it change

6. Dynamic Auscultation

- Does inspiration worsen it (L vs R)

7. Clinical Signs

- Does this patient present with any signs of left or right HF, signs of other clinical manifestations

Algorithm: Exam Questions

- Board questions now will include audio sounds for murmur questions. You must be able to use both the question stem and the actual sound of the murmur to answer these questions
- Typically, the case presentation should point you in the right direction
 - Systolic vs diastolic
 - Location
 - Signs of heart failure (left vs right vs both)
 - Dynamic auscultation
 - Patient History
 - Hx of untreated Strep throat: think mitral valve issues (rheumatic heart disease)
 - Elderly individual with dyspnea, radiation to carotids: aortic stenosis
 - Pillow orthopnea: left sided heart failure
- Have an idea of what the murmur should be before you listen to the audio

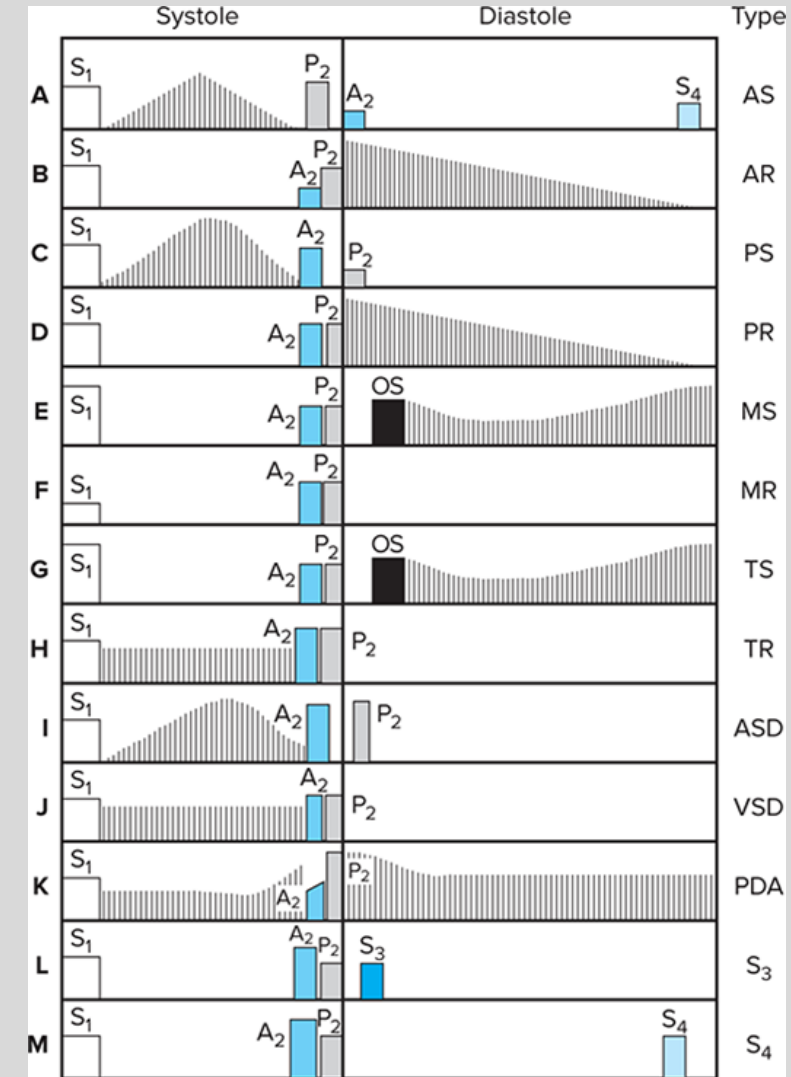
Summary

Link to heart sounds:

- <https://depts.washington.edu/physdx/heart/demo.html>

Dynamic Auscultation

- Inspiration worsens R-sided murmurs
- Hand grip worsens L-sided regurgitations
- Rapid squatting and leg raise $\uparrow$ preload
- Standing and Valsalva $\downarrow$ preload
- Maneuvers that $\downarrow$ preload improve most murmurs
- Maneuvers that $\uparrow$ preload worsen most murmurs
- Exceptions: MVP and HOCM



Source: Leonard G. Gomella, Steven A. Haist: *Gomella and Haist's Clinician's Pocket Reference, 12e*
Copyright © Leonard G. Gomella. All rights reserved.

Systolic Murmur	Auscultation*	Other clinical findings	Cause
Aortic Stenosis	S: crescendo-decrescendo L/R: R 2 nd ICS near sternal border, radiates to carotids Q: harsh I: loud P: medium	↑ intensity with squatting, leaning forward, expiration - Reversed splitting of S₂ - Syncope, angina, and dyspnea on exertion - Pulsus parvus et tardus	Calcified valve: age-related degeneration or congenital bicuspid aortic valve
Pulmonic Stenosis	S: crescendo-decrescendo L/R: L 2 nd ICS near sternal border, radiates to L shoulder and neck Q: harsh I: soft to loud P: medium	↑ intensity with inspiration - Pathological splitting of S₂	Congenital
Mitral Regurgitation	S: holosystolic L/R: apex, radiates to L axilla Q: blowing I: soft to loud P: high	↑ intensity with handgrip and expiration	- Mitral valve prolapse - Papillary muscle rupture (post-MI) - LV dilation
Mitral Valve Prolapse	S: Late crescendo, preceded by midsystolic click L: apex	↑ intensity and delayed MC with squatting - Typically asymptomatic - Can lead to MR	Myxomatous degeneration
Tricuspid Regurgitation	S: holosystolic L/R: L lower sternal border, radiates to R of the sternum Q: blowing I: variable P: medium	↑ intensity with inspiration - Associated with increased pressure in pulmonary arteries	- Ebstein anomaly - Infective endocarditis in IV drug user

Diastolic Murmur	Auscultation*	Other clinical findings	Cause
Aortic Regurgitation	S: early decrescendo L/R: L midsternal border, radiates to the apex Q: blowing I: soft to loud P: high	• ↑ intensity with handgrip, leaning forward, expiration • Widened pulse pressure • Head bobbing • Bounding pulse	• Congenital bicuspid aortic valve • Aortic root dilation
Pulmonic Regurgitation	S: early decrescendo L/R: L 2 nd ICS near sternal border, radiates along L sternal border Q: blowing I: soft P: high	• ↑ intensity with inspiration	• Pulmonary hypertension
Mitral Stenosis	S: Decrescendo following opening snap L/R: apex, non-radiating Q: rumbling I: soft to loud P: low	• ↑ intensity with expiration, left lateral decubitus	• Chronic rheumatic heart disease
Tricuspid Stenosis	S: Decrescendo following opening snap L/R: L lower sternal border, non-radiating Q: rumbling I: soft P: low	• ↑ intensity with inspiration • Rare	

Question 1

A 75-year-old man comes to the clinic because of dizziness and shortness of breath with exertion. During these times, he feels generalized chest discomfort that is relieved with rest. He has a history of hypertension that is well controlled with enalapril. Physical examination discloses a loud, mid-to-late systolic murmur in the right upper sternal border radiating to the carotids.

Which of the following is most likely also to be seen in this patient?

- A. A decrease in murmur intensity with Valsalva maneuver
- B. A decrease in murmur intensity with squatting
- C. An increase in murmur intensity with handgrip
- D. An increase in murmur intensity with inspiration
- E. Head bobbing
- F. Hoarseness

Question 1: Answer

A 75-year-old man comes to the clinic because of dizziness and shortness of breath with exertion. During these times, he feels generalized chest discomfort that is relieved with rest. He has a history of hypertension that is well controlled with enalapril. Physical examination discloses a loud, mid-to-late systolic murmur in the right upper sternal border radiating to the carotids.

Which of the following is most likely also to be seen in this patient?

- A. **A decrease in murmur intensity with Valsalva maneuver**
- B. A decrease in murmur intensity with squatting
- C. An increase in murmur intensity with handgrip
- D. An increase in murmur intensity with inspiration
- E. Head bobbing
- F. Hoarseness

Explanation: This patient is suffering from aortic stenosis. Keys to identifying this are the timing of the murmur (loud, mid-to late systolic murmur) and the location (RUS; Aortic Region), and the characteristics (radiation to the carotids). Since aortic stenosis is a forward flowing murmur, we would expect anything that decreases the forward flow of blood to decrease intensity of the murmur (decrease preload or increased afterload).

Question 2

A 40-year-old man comes to the emergency department with acute-onset weakness on his left side. He also reports that his right calf has been swollen for the past 2 days, following a long flight. He denies dyspnea or chest pain. A CT scan shows a likely embolic stroke in the right middle cerebral artery distribution.

Which of the following is most likely also to be seen in this patient?

- A. A rumbling mid-diastolic murmur best heard at the apex
- B. Continuous machine-like murmur and bounding peripheral pulses
- C. Low-frequency, vibratory ejection systolic murmur at the left lower sternal border, near the apex
- D. Pansystolic vibratory murmur
- E. Wide and fixed split S2

Question 2: Answer

A 40-year-old man comes to the emergency department with acute-onset weakness on his left side. He also reports that his right calf has been swollen for the past 2 days, following a long flight. He denies dyspnea or chest pain. A CT scan shows a likely embolic stroke in the right middle cerebral artery distribution.

Which of the following is most likely also to be seen in this patient?

- A. A rumbling mid-diastolic murmur best heard at the apex
- B. Continuous machine-like murmur and bounding peripheral pulses
- C. Low-frequency, vibratory ejection systolic murmur at the left lower sternal border, near the apex
- D. Pansystolic vibratory murmur
- E. **Wide and fixed split S2**

Explanation: This patient has had an embolic stroke in the Right MCA, with a previous history of DVTs. The only way for the emboli to travel from the venous system to the arterial system without passing through the lungs (which results in a PE) is through a hole in the atria or ventricles. ASDs typically create a split S2, as increased preload is present in the RV

Question 3

A 43-year-old woman comes to the clinic because of increasing shortness of breath during activities for the past several months. She reports having to catch her breath after walking up a flight of stairs and needing to take frequent breaks while performing chores. She is an immigrant from a developing country and moved to the United States 10 years ago. Hemodynamic studies show abnormally elevated pulmonary capillary wedge pressure with normal left ventricular pressure.

Which of the following mechanisms underlies this patient's most likely condition?

- A. Age-related valvular calcification
- B. Congenital defect
- C. Molecular mimicry
- D. Myocardial ischemia
- E. Papillary muscle rupture
- F. Pulmonary arterial thrombi

Question 3: Answer

A 43-year-old woman comes to the clinic because of increasing shortness of breath during activities for the past several months. She reports having to catch her breath after walking up a flight of stairs and needing to take frequent breaks while performing chores. She is an immigrant from a developing country and moved to the United States 10 years ago. Hemodynamic studies show abnormally elevated pulmonary capillary wedge pressure with normal left ventricular pressure.

Which of the following mechanisms underlies this patient's most likely condition?

- A. Age-related valvular calcification
- B. Congenital defect
- C. Molecular mimicry**
- D. Myocardial ischemia
- E. Papillary muscle rupture
- F. Pulmonary arterial thrombi

Explanation: This patient has mitral stenosis, most likely from rheumatic fever during childhood. She has elevated pulmonary vasculature, suggesting an issue on the left side of the heart (increases pressure in LA due stenotic valve). Mitral stenosis is typically due to rheumatic fever, where antibodies interact with the antigen of GAS and the mitral valve leaflets.

Question 4

A 43-year-old woman comes to the clinic because of increasing shortness of breath during activities for the past several months. She reports having to catch her breath after walking up a flight of stairs and needing to take frequent breaks while performing chores. She is an immigrant from a developing country and moved to the United States 10 years ago. Hemodynamic studies show abnormally elevated pulmonary capillary wedge pressure with normal left ventricular pressure.

Which of the following best describes the heart murmur that this patient most likely has?

- A. Blowing diastolic murmur heard best at the left sternal border
- B. Harsh crescendo decrescendo ejection murmur
- C. Machine line murmur
- D. Mid-diastolic rumble with an opening snap at the apex
- E. Systolic murmur at the right sternal border

Question 4: Answer

A 43-year-old woman comes to the clinic because of increasing shortness of breath during activities for the past several months. She reports having to catch her breath after walking up a flight of stairs and needing to take frequent breaks while performing chores. She is an immigrant from a developing country and moved to the United States 10 years ago. Hemodynamic studies show abnormally elevated pulmonary capillary wedge pressure with normal left ventricular pressure.

Which of the following best describes the heart murmur that this patient most likely has?

- A. Blowing diastolic murmur heard best at the left sternal border (Pulmonic regurgitation)
- B. Harsh crescendo decrescendo ejection murmur (Pulmonary valve stenosis)
- C. Machine line murmur (Patent ductus arteriosus)
- D. Mid-diastolic rumble with an opening snap at the apex**
- E. Systolic murmur at the right sternal border (Aortic stenosis)

Explanation: Mitral stenosis will present with a mid-diastolic rumble with an opening snap at the apex.

Question 5

A 23-year-old man who injects heroin is brought to the hospital by paramedics after being found unconscious and resuscitated with naloxone. His temperature is 39°C (102.2°F). Upon examination, there is a prominent jugular venous pulse and a “blowing” systolic heart murmur loudest at the left lower sternal border. The murmur varies with respiration, becoming louder with inspiration.

Which of the following is the most likely cardiac diagnosis in this patient?

- A. Aortic regurgitation
- B. Aortic stenosis
- C. Mitral regurgitation
- D. Mitral stenosis
- E. Tricuspid regurgitation

Question 5: Answer

A 23-year-old man who injects heroin is brought to the hospital by paramedics after being found unconscious and resuscitated with naloxone. His temperature is 39°C (102.2°F). Upon examination, there is a prominent jugular venous pulse and a “blowing” systolic heart murmur loudest at the left lower sternal border. The murmur varies with respiration, becoming louder with inspiration.

Which of the following is the most likely cardiac diagnosis in this patient?

- A. Aortic regurgitation
- B. Aortic stenosis
- C. Mitral regurgitation
- D. Mitral stenosis
- E. Tricuspid regurgitation**
- F. Tricuspid stenosis

Explanation: Patient has a murmur that increases with inspiration, and is a systolic murmur. This leaves only pulmonary valve stenosis and tricuspid regurgitation as options. Patient is an IV drug user, which commonly causes infective endocarditis of the tricuspid valve.

Question 6

A 21-year-old female comes to your office with complaint of feeling “extra beats” in her chest. She has no history of syncope or presyncope, no cardiovascular risk factors and denies a family history of cardiovascular disease. She has no signs or symptoms of congestive heart failure and takes no medications. On physical exam, her vital signs are normal. Her lungs are clear. A grade 3/6 late systolic murmur is present. It is heard best at the cardiac apex. A mid systolic click is heard, and echocardiogram confirms diagnosis of mitral valve prolapse.

Which of the following maneuvers will **INCREASE** the intensity of the murmur and make the click occur **LATER** in systole

- A. Valsalva maneuver
- B. Standing
- C. Inspiration
- D. Leg raise
- E. Handgrip

Question 6: Answer

A 21-year-old female comes to your office with complaint of feeling “extra beats” in her chest. She has no history of syncope or presyncope, no cardiovascular risk factors and denies a family history of cardiovascular disease. She has no signs or symptoms of congestive heart failure and takes no medications. On physical exam, her vital signs are normal. Her lungs are clear. A grade 3/6 late systolic murmur is present. It is heard best at the cardiac apex. A mid systolic click is heard, and echocardiogram confirms diagnosis of mitral valve prolapse.

Which of the following maneuvers will **DECREASE** the intensity of the murmur and make the click occur **LATER** in systole

- A. Valsalva maneuver
- B. Standing
- C. Inspiration
- D. Leg raise**
- E. Handgrip

Explanation: Mitral valve prolapse occurs during systole, and events that increase the preload will decrease the intensity of the murmur and move the click of MVP later in systole. All other options either decrease preload or increase afterload on the left side of the heart.

Thank You!

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine